Recommended Assessment

Microphone & Speaker

Exploring Speaker

1. Why can a sine wave be used to characterize the speaker's performance?

2. As you increased the sine wave frequency, how did the amplitude output from the speaker change, and what property of the speaker explains this?

Microphone Visualization and Application

3. Why does changing the sample rate affect the quality of recorded data?

4. What is the significance of block size when processing real-time audio data?

5. Why is it useful to analyze microphone data in the frequency domain (using FFT)?

Fast Fourier Transform (FFT) and Frequency Analysis

6. Why does FFT block size affect the accuracy and responsiveness of frequency detection?

7. Why the FFT curve often shows a large magnitude near 0 Hz, even when no low-frequency tone is intentionally present.

8. When you implemented Clap detection, how does it respond to sound sources? How can you improve its reliability and isolate claps better?

Microphone & Speaker Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	